COURSE

Partial exam: 3OP { Theory 1, Theory 2 } MIN ≥ 5; { Problem 1, Problem 2 } MIN ≥ 5

Course: 1OP (prusenca + answers)

Seminar: 1OP (prusenca + activity)

Final exam: 5OP

PART I SYSTEMS OF DIFFERENTIAL EQUATIONS (SDEs)

CHAPTER I LINEAR SDEs

INTRODUCTION

A linear SDE has the form

$$\begin{cases} y_1'(x) = a_{11}(x)y_1(x) + a_{12}(x)y_2(x) + \dots + a_{1n}(x)y_n(x) + b_1(x) \\ y_2'(x) = a_{21}(x)y_1(x) + a_{22}(x)y_2(x) + \dots + a_{2n}(x)y_n(x) + b_2(x) \\ \dots \\ y_n'(x) = a_{n1}(x)y_1(x) + a_{n2}(x)y_2(x) + \dots + a_{nn}(x)y_n(x) + b_n(x) \end{cases} \quad (1)$$

$a_{ij} : I \longrightarrow \mathbb{R}$

$I \subset \mathbb{R}$ interval

a_{ij} and b_j are continous functions on the interval I

- we can show that the solution exists

The functions $b_1, b_2, \dots, b_n$ are called perturbing functions

The system (1) is said to be nonhomogeneous if $b_1, b_2, \dots, b_n \equiv 0$ is said to be homogeneous

introduce the matrix and vectors

$$A(x) = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

$$y(x) = \begin{pmatrix} y_1(x) \\ y_2(x) \\ \vdots \\ y_n(x) \end{pmatrix}$$

$$b(x) = \begin{pmatrix} b_1(x) \\ b_2(x) \\ \vdots \\ b_n(x) \end{pmatrix}$$

The equations (1) are equivalent to the equality of the vectors $y'(x)$ and $A(x) \cdot y(x) + b(x)$

$\Rightarrow$ A nonhomogeneous SDE has the form

$$y'(x) = A(x) \cdot y(x) + b(x) \quad (1)$$

A homogeneous SDE has the form

$$y'(x) = A(x) \cdot y(x) \quad (1)$$

A solution is a vector function $y(x) = \begin{pmatrix} y_1(x) \\ \vdots \\ y_n(x) \end{pmatrix}$ which satisfy SDE (1) or (1')

The general solution (of (1) or (1')) is a vector ~~function~~ function $y(x, c_1, c_2, \dots, c_n)$ where $c_1, c_2, \dots, c_n \in \mathbb{R}$ are arbitrary constants such that, any solution can be ~~arbitrary~~ obtained from this $y(\dots)$ by a suitable choice of the arbitrary constants

- variable x represents the time

An initial value problem (IVP) is formed by

$$\begin{cases} y'(x) = A(x)\, y(x) + b(x) \quad (1) \text{ SDE} \\ y(x_0) = y_0 \quad \text{(initial condition)} \end{cases}$$

where $x_0 \in \mathbb{R}$ is the initial moment and $y_0 \in \mathbb{R}^n$ is the initial position

S1. Linear and homogeneous SDE's

$$y'(x) = A(x)\,y(x) \quad (1')$$

Proposition 1: Any linear combination of solutions of $(1')$ is a solution of the SDE $(1')$

Proof: Let $y_1(x)$, $y_2(x)$ two solutions of the SDE $(1')$

$$\Rightarrow y_1'(x) = A(x)\,y_1(x), \quad y_2'(x) = A(x)\cdot y_2(x) \quad (2)$$

Let c_1, c_2 two real constants

The function $y(x) = c_1 y_1(x) + c_2 y_2(x)$ is a linear combination of $y_1(x)$ and $y_2(x)$

Introduce $y(x)$ in $(1')$

$$y'(x) = c_1 y_1'(x) + c_2 y_2'(x) \overset{(2)}{=} c_1 A(x)\,y_1(x) + c_2 A(x)\,y_2(x)$$

Take $A(x)$ as a left common factor $\Rightarrow A(x)\,(c_1 y_1(x) + c_2 y_2(x))$

$$= A(x)\,y(x)$$

$\Rightarrow y'(x) = A(x)\,y(x) \Rightarrow$ indeed the linear combination $y(x)$ is a solution of $(1')$

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Corollary: The set of solutions of a homogeneous SDE is a linear space

Wronskian

Consider n solutions of the SDE (1)

$$y_1 = \begin{pmatrix} y_{11} \\ y_{12} \\ \vdots \\ y_{1n} \end{pmatrix}, \quad y_2 = \begin{pmatrix} y_{21} \\ y_{22} \\ \vdots \\ y_{2n} \end{pmatrix}, \dots, \quad y_n = \begin{pmatrix} y_{n1} \\ y_{n2} \\ \vdots \\ y_{nn} \end{pmatrix}$$

Definition: The determinant formed with the columns $y_1, y_2, \dots, y_n$ is called the Wronskian of the solutions

Proposition 2: If the solutions $y_1, y_2, \dots, y_n$ are linearly dependent on an interval $I \subset \mathbb{R}$ then ~~$W(x) = 0_n, \forall x \in I$~~ $W(x) = 0, \forall x \in I$

Proof: Consider a chain equivalent statements:

The solutions $y_1, y_2, \dots, y_n$ are linearly dependent on I
$\Rightarrow \exists\, c_1, c_2, \dots, c_n$ not all equal to 0, such that

$$c_1 y_1(x) + c_2 y_2(x) + \dots + c_n y_n(x) = \overline{0}, \quad \forall x \in I$$

$$\Rightarrow \begin{cases} C_1 y_{11} + C_2 y_{12} + \dots + C_n y_{1n} = 0 \\ C_1 y_{21} + C_2 y_{22} + \dots + C_n y_{2n} = 0 \\ \vdots \\ C_1 y_{n1} + C_2 y_{n2} + \dots + C_n y_{nn} = 0 \end{cases}$$

$\Rightarrow$ consider these equations as an algebric linear system which has the solution $(C_1, C_2, \dots, C_n)$

The algebric system is linear and homogeneous and has a nonzero solution $(C_1, C_2, \dots, C_n) \Rightarrow$ the determinant of this system $= 0 \Rightarrow$

$$\Rightarrow \begin{vmatrix} y_{11} & y_{12} & \dots & y_{1n} \\ y_{21} & y_{22} & \dots & y_{2n} \\ \vdots & & & \\ y_{n1} & y_{n2} & \dots & y_{nn} \end{vmatrix} = 0 = W(x)$$

$W(x) = 0$

By negation we can prove

Proposition 3: The solution $y_1, y_2, \dots, y_n$ of the homogeneous SDE (1) are linearly independent on $I \Leftrightarrow$ $W(x) \neq 0$

In this case ($W(x) \neq 0$) the solutions $y_1, y_2, \dots, y_n$ form a basis in the linear space of solutions $\Leftrightarrow$ any solution is a linear combination of the basis

$\Rightarrow$ any solution has the form

$y(x) = C_1 y_1 + C_2 y_2 + \dots + C_n y_n$ (3) $\Rightarrow$ this is the general solution

Therefore $W(x) \neq 0$ then (3) is the general solution

S2. Linear and homogeneous SDEs with constants coefficients

All coefficients $a_{ij}(x) = a_{ij} \in \mathbb{R}$ constants $\Rightarrow$

$\Rightarrow$ the matrix $A = (a_{ij})$ is a constant matrix

Case I: The matrix A is diagonalizable $\Leftrightarrow$ the matrix A has n linearly independent eigenvectors $V_1, V_2, \dots, V_n$ corresponding to n eigenvalues $\Rightarrow$

$$\Rightarrow \begin{aligned} A V_1 &= \lambda_1 V_1 \\ A V_2 &= \lambda_2 V_2 \\ &\dots \\ A V_n &= \lambda_n V_n \end{aligned} \quad (2)$$

linearly independent $\Rightarrow$ the determinant $|V_1, V_2 \dots V_n| \neq 0$

Then the vector functions

$y_1 = V_1 e^{\lambda_1 x}, \dots, y_n = V_1 e^{\lambda_n x} + \dots + V_n e^{\lambda_n x}$ (3)

are linearly independent solutions of

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Linear and Homogeneous SDEs

$Y' = A \cdot Y$, $A \in \mathcal{M}_n(\mathbb{R})$, $Y = \begin{pmatrix} y_1(x) \\ y_2(x) \\ \vdots \\ y_n(x) \end{pmatrix}$ - the unknown

Algorithm:

1. Find the eigenvalues of A ($\det(A - \lambda I_n) = 0$)
Find $\overline{V}(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$ and their algebric multiplicities $ma(\lambda_1)$; $ma(\lambda_2)$ ....

2. For every $\lambda \in \overline{V}(A)$, we need the eigenspace and the geometric multiplicity of λ ($(A - \lambda I_n) \cdot V = 0 \Rightarrow$ eigenspace
$mg(\lambda)$ = the dimension of the eigenspace)

3. For every $\lambda \in \overline{V}(A)$

a. If $ma(\lambda) = mg(\lambda) = 1 \Rightarrow$ take V eigenvector of λ and create $e^{\lambda x} \cdot V$ solution of SDE

b. If $ma(\lambda) = mg(\lambda) = K \geq 2 \Rightarrow$ take $V_1, V_2, \dots, V_K$ linearly independent vectors

c. If $ma(\lambda) > mg(\lambda)$, but $mg(\lambda) = 1 \Rightarrow$ take V eigenvectors of λ and create principle vectors $V_{p_1}, V_{p_2}, \dots, V_{ma(\lambda)-1}$

$(A - \lambda I_n) V_{p_1} = V$

$(A - \lambda I_n) V_{p_2} = V_{p_1}$

$V \rightarrow$ solution $e^{\lambda x} \cdot V$

$V_{p_1} \rightarrow e^{\lambda x}\left(\frac{x}{1!} \cdot V + V_{p_1}\right)$

$V_{p_2} \rightarrow e^{\lambda x}\left(\frac{x^2}{2!} \cdot V + \frac{x}{1!} V_{p_1} + V_{p_2}\right)$

d. If $ma(\lambda) > mg(\lambda)$, but $mg(\lambda) = k$

4. The general solution

$$Y = C_1\left(\begin{matrix}1^{st}\text{ set of}\\ \text{SDE from}\\ \text{step 3}\end{matrix}\right) + C_2\left(\begin{matrix}2^{nd}\text{ set of}\\ \text{SDE from}\\ \text{step 3}\end{matrix}\right) + \dots + C_n\left(\begin{matrix}\text{last set of}\\ \text{SDE from}\\ \text{step 3}\end{matrix}\right)$$

1. $Y' = A \cdot Y$, $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 0 \\ 1 & 2 & 0 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I_3) = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 2 & 0 \\ 0 & -1-\lambda & 0 \\ 1 & 2 & -\lambda \end{vmatrix} = 0$$

$$-\lambda \begin{vmatrix} 1-\lambda & 2 \\ 0 & -1-\lambda \end{vmatrix} = -\lambda(1-\lambda)^2 = 0 \Rightarrow \lambda_1 = 0$$

$$\lambda_2 = \lambda_3 = 1$$

$ma(\lambda_1) = 1$

$ma(\lambda_2) = 2$

$V(A) = \{0, 1\}$

$\lambda=0 \Rightarrow (A-\lambda I_3)\cdot V=0 \Rightarrow \begin{pmatrix}1&2&0\\0&1&0\\1&2&0\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$

$\begin{cases}a+2b=0\\b=0\\a+2b=0\end{cases} \Rightarrow \begin{matrix}a=0\\b=0\\c\in\mathbb{R}\end{matrix}$

$\begin{pmatrix}a\\b\\c\end{pmatrix}=\begin{pmatrix}0\\0\\c\end{pmatrix} \to$ ~~$mg\,t=1$~~ $mg=1 \Rightarrow c\begin{pmatrix}0\\0\\1\end{pmatrix}$

$\lambda_2=1 \Rightarrow (A-\lambda_2 I_3)\cdot V=0 \Rightarrow \begin{pmatrix}0&2&0\\0&0&0\\1&2&-1\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$

$\begin{cases}2b=0\\a+2b-c=0 \Rightarrow a=c\end{cases}$

$\begin{pmatrix}a\\b\\c\end{pmatrix}=\begin{pmatrix}c\\0\\c\end{pmatrix} \to$ ~~$mg\,t$~~ $mg=1 \Rightarrow c\begin{pmatrix}1\\0\\1\end{pmatrix}$

$\lambda_1=0,\ ma=1,\ mg=1 \Rightarrow y_1=e^{0\cdot x}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}0\\0\\1\end{pmatrix}$

$\lambda_2=1,\ ma=2,\ mg=1 \Rightarrow (A-\lambda I_3)V_{p_1}=V$

$\begin{pmatrix}0&2&0\\0&0&0\\1&2&-1\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix}=\begin{pmatrix}1\\0\\1\end{pmatrix} \Rightarrow \begin{cases}2b=0 \Rightarrow b=\frac{1}{2}\\a+2b-c=1 \Rightarrow a=c\end{cases}$

$V_{p_1}=\begin{pmatrix}c\\\frac{1}{2}\\c\end{pmatrix}=c\begin{pmatrix}1\\0\\1\end{pmatrix}+\underbrace{\begin{pmatrix}0\\1/2\\0\end{pmatrix}}_{V_{p_1}}$

$$\begin{pmatrix}1\\0\\1\end{pmatrix} \longrightarrow Y_2 = e^{1x}\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}e^x\\0\\e^x\end{pmatrix}$$

$$V_{\varphi_1} = \begin{pmatrix}0\\\frac{1}{2}\\0\end{pmatrix} \longrightarrow Y_3 = e^{1x}\left(\frac{x}{1!}\cdot\begin{pmatrix}1\\0\\1\end{pmatrix} + \begin{pmatrix}0\\\frac{1}{2}\\0\end{pmatrix}\right) = \begin{pmatrix}x e^x\\e^x/2\\x e^x\end{pmatrix}$$

$$Y = C_1\cdot\begin{pmatrix}0\\0\\1\end{pmatrix} + C_2\begin{pmatrix}e^x\\0\\e^x\end{pmatrix} + C_3\begin{pmatrix}x e^x\\e^x/2\\x e^x\end{pmatrix},\ C_1, C_2, C_3 \in \mathbb{R}$$

2. $Y' = A\cdot Y,\ A = \begin{pmatrix}1 & -2 & -2\\-2 & 1 & -2\\-2 & -2 & 1\end{pmatrix}$

$$P(\lambda) = |A - \lambda I_3| = 0 \Rightarrow \begin{vmatrix}1-\lambda & -2 & -2\\-2 & 1-\lambda & -2\\-2 & -2 & 1-\lambda\end{vmatrix} = 0$$

$$(-3-\lambda)\begin{vmatrix}1 & 1 & 1\\2 & 1-\lambda & -2\\-2 & -2 & 1-\lambda\end{vmatrix} = 0$$

$$(-3-\lambda)\begin{vmatrix}1 & 0 & 0\\-2 & 3-\lambda & 0\\-2 & 0 & 3-\lambda\end{vmatrix} = 0$$

$$(-3-\lambda)\begin{vmatrix}3-\lambda & 0\\0 & 3-\lambda\end{vmatrix} = -(3+\lambda)(3-\lambda)^2 = 0 \Rightarrow \lambda_1 = -3$$
$$\lambda_2 = \lambda_3 = 3$$

$$\lambda_1 = -3 \Rightarrow (A - \lambda_1 I_3)\cdot V = 0 \Rightarrow \begin{pmatrix}4 & -2 & -2\\-2 & 4 & -2\\-2 & -2 & 4\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

$$\begin{cases}4x - 2y - 2z = 0 \Rightarrow z = 2x + y = 5x\\-2x + 4y - 2z = 0 \Rightarrow -2x + 4y - 4x - 2y \Rightarrow -6x + 2y \Rightarrow y = 3x\\-2x - 2y + 4z = 0\end{cases}$$

$$\begin{cases} 4x-2y-2z=0 \\ -2x+4y-2z=0 \\ -2x-2y+4z=0 \end{cases}$$

$$\begin{cases} 4x-2y-2z=0 \\ -2x+4y-2z=0 \end{cases} \Rightarrow 6x-6y=0 \Rightarrow x=y$$

$$4y-2y-2z=0 \Rightarrow 2y-2z=0 \Rightarrow z=y$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} y \\ y \\ y \end{pmatrix} = y\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \rightarrow mg(\lambda_1)=1$$

$$\lambda_2=3 \Rightarrow (A-\lambda_2 I_3)\cdot V=0 \Rightarrow \begin{pmatrix} -2 & -2 & -2 \\ -2 & -2 & -2 \\ -2 & -2 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} -2x-2y-2z=0 \Rightarrow z=-x-y \\ -2x-2y-2z=0 \\ -2x-2y-2z=0 \end{cases}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ -x-y \end{pmatrix} = \begin{pmatrix} x \\ 0 \\ -x \end{pmatrix} + \begin{pmatrix} 0 \\ y \\ -y \end{pmatrix} \rightarrow mg(\lambda_2)=2$$

$$\lambda_1=-3,\ ma=1,\ mg=1 \Rightarrow y_1 = e^{-3x}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} e^{-3x} \\ e^{-3x} \\ e^{-3x} \end{pmatrix}$$

$$\lambda_2=3,\ ma=2,\ mg=2 \Rightarrow y_2 = e^{3x}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} e^{3x} \\ 0 \\ -e^{3x} \end{pmatrix}$$

$$y_3 = e^{3x}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ e^{3x} \\ -e^{3x} \end{pmatrix}$$

$$y = C_1\begin{pmatrix} e^{-3x} \\ e^{-3x} \\ e^{-3x} \end{pmatrix} + C_2\begin{pmatrix} e^{3x} \\ 0 \\ -e^{3x} \end{pmatrix} + C_3\begin{pmatrix} 0 \\ e^{3x} \\ -e^{3x} \end{pmatrix},\ C_1, C_2, C_3 \in \mathbb{R}$$

2. $Y' = A \cdot Y$, $A = \begin{pmatrix} 2 & -1 & 2 \\ 5 & -3 & 3 \\ -1 & 0 & -2 \end{pmatrix}$

$P(\lambda) = \det(A - \lambda I_3) = 0 \Rightarrow \begin{vmatrix} 2-\lambda & -1 & 2 \\ 5 & -3-\lambda & 3 \\ -1 & 0 & -2-\lambda \end{vmatrix} = 0$

$$\begin{vmatrix} 2-\lambda & -1 & 2 \\ 5 & -3-\lambda & 3 \\ -1 & 0 & -2-\lambda \end{vmatrix} = 0$$

$2-\lambda \quad -1 \quad 2$

$5 \quad -3-\lambda \quad 3$

$\begin{vmatrix} 2-\lambda & -1 & 2 \\ 5 & -3-\lambda & 3 \\ -1 & 0 & -2-\lambda \end{vmatrix} = 0 \Rightarrow (2-\lambda)\begin{vmatrix} -3-\lambda & 3 \\ 0 & -2-\lambda \end{vmatrix} + 1\begin{vmatrix} 5 & 3 \\ -1 & -2-\lambda \end{vmatrix} + 2\begin{vmatrix} 5 & -3-\lambda \\ -1 & 0 \end{vmatrix}$

$= (2-\lambda)(-3-\lambda)(-2-\lambda) + 10 - 5\lambda + 3 - 6 - 2\lambda$

$+ (4-\lambda^2)(+3+\lambda) - 10 - 7\lambda$

$12 + 4\lambda - 3\lambda^2 - \lambda^3 - 10 - 7\lambda$

~~$-\lambda^3 - 3\lambda^2 - 3\lambda + 2 = 0$~~ $\quad -\lambda^3 - 3\lambda^2 - 3\lambda - 1 = 0$

$(\lambda+1)(-\lambda^2 - 2\lambda - 1) = 0$

$\rightarrow \lambda = -1$

$\rightarrow -(\lambda+1)^2 = 0 \rightarrow \lambda = -1$

$\lambda = -1, \quad m_a = 3$

$-\lambda^3 - 3\lambda^2 - 3\lambda - 1 \;\big|\; \lambda + 1$

$+\lambda^3 + \lambda^2 \qquad\qquad -\lambda^2 - 2\lambda - 1$

$-2\lambda^2 - 3\lambda$

$+2\lambda + 2\lambda$

$-\lambda - 1$

$ax^3+bx^2+cx+d=0$

	a	b	c	d
λ	a	$\lambda\cdot a+b$ (A)	$\lambda A+c$ (B)	$\lambda B+d$

$\neq 0 \Rightarrow \lambda$ is not a root

$= 0 \Rightarrow \lambda$ is a root

$$\lambda=-1 \Rightarrow (A-\lambda I_3)\cdot V=0 \Rightarrow \begin{pmatrix}3 & -1 & 2\\ 5 & -2 & 3\\ -1 & 0 & -1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

$$\begin{cases}3x-y+2z=0 \Rightarrow -3z-y+2z=0 \Rightarrow -y=z \Rightarrow y=-z\\ 5x-2y+3z=0\\ -x-z=0 \Rightarrow x=-z\end{cases}$$

$$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}z\\-z\\z\end{pmatrix}=z\begin{pmatrix}1\\-1\\1\end{pmatrix} \Rightarrow mg=1$$

$$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-z\\-z\\z\end{pmatrix}=\begin{pmatrix}-1\\-1\\1\end{pmatrix} \Rightarrow mg=1$$

We have to find v_{p_1}, v_{p_2}

$$(A-\lambda I_3)\,v_{p_1}=V \Rightarrow \begin{pmatrix}3 & -1 & 2\\ 5 & -2 & 3\\ -1 & 0 & -1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-1\\-1\\1\end{pmatrix}$$

$$\begin{cases}3x-y+2z=-1 \Rightarrow -3z-3-y+2z=-1 \Rightarrow -z-3-y=-1\\ 5x-2y+3z=-1\\ -x-z=1 \Rightarrow x=-z-1\end{cases}$$

$$-y=2-z \Rightarrow y=z-2$$

$$\begin{pmatrix}-z-1\\-z-2\\z\end{pmatrix}=\begin{pmatrix}-z\\-z\\z\end{pmatrix}+\underbrace{\begin{pmatrix}-1\\-2\\0\end{pmatrix}}_{v_{p_1}}$$

$(A-\lambda I_3)\cdot V_{p_2} = V_{p_1}$

$$\begin{pmatrix} 3 & -1 & 2 \\ 5 & -2 & 3 \\ -1 & 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix}$$

$3x - y + 2z = -1$

$5x - 2y + 3z = -2 \Rightarrow$

$-x - z = 0 \Rightarrow x = -z$

COURSE

Systems of differential equations SDE's linear, homogeneous with constant coefficient

$Y'(x) = A \cdot Y(x)$ (1) A an $n \times n$ constant matrix

Algorithm

Step I: Solve the characteristic equation of A and determine the eigenvalues and determine the ~~spectrum~~ spectrum of A: $G(A) = \{\lambda_1, \lambda_2 \dots\}$

Step II: For each eigenvalue λ_i Solve the algebraic linear and homogeneous system ~~$(A - I_n)\cdot V = \bar{0}$~~ $(A - \lambda I_n)\cdot V = 0$ and determine a corresponding eigenvector V_i

Step III: Write the general solution of (1')

$$y(x) = C_1 \cdot v_1 e^{\lambda_1 x} + C_2 v_2 e^{\lambda_2 x} + \dots + C_n v_n e^{\lambda_n x}$$

Case II: The matrix A is not diagonalizable
(when A has not n linearly independent eigenvalues)

Let λ be an eigenvalue of the matrix A i.e a root of the characteristic equation ~~$\det(\lambda I$~~ $\det(A - \lambda I) = 0$

We denote by $m_a(x)$ the algebraic multiplicitude multitude of λ = the multitude of λ as root of the characteristic equation

$$A = \begin{vmatrix} -1 & 0 & 2 \\ 0 & -1 & 3 \\ 0 & 0 & 5 \end{vmatrix}$$

$\det(\lambda I - A) = (\lambda+1)^2(\lambda - 5) \Rightarrow m_a(-1) = 2$

$m_a(5) = 1$

The geometric multiplicity of λ $m_g(\lambda)$ = number of linearly independent ~~eigenvalues~~ eigenvectors corresponding to the eigenvalue λ $m_a(\lambda) \geq m_g(\lambda)$

v_1 for λ_1 => generalised eigenvector $Av_2 = \lambda v_2$

v_2 for λ_2 $\qquad Av_3 = \lambda v_3 + v_1$

Case II => $\exists\ \lambda \in G(A)$ (λ eigenvalue of A) with $ma(\lambda) > mg(\lambda)$

=> not enough linearly independent eigenvectors =>

=> introduce chains of generalised eigenvectors

$Av_1 = \lambda v_1$, $v_1 \neq \bar{0}$ (v_1 is eigenvector)

$Av_2 = \lambda v_2 + v_1$

$Av_3 = \lambda v_3 + v_2$

$\dots\dots\dots\dots$

$Av_m = \lambda v_m + v_{m-1}$

Write the corresponding linearly independent solutions of SDE (1')

$y_1 = v_1 e^{\lambda x}$ -> 0 degree polynomial

$y_2 = \left[\frac{x}{1!} \cdot v_1 + v_2\right] e^{\lambda x}$ -> 1 degree polynomial

$y_3 = \left[\frac{x^2}{2!} \cdot v_1 + \frac{x}{1!} \cdot v_2 + v_3\right] e^{\lambda x}$ -> 2 degree polynomial (3)

$\dots\dots\dots\dots$

$y_m = \left[\frac{x^{m-1}}{(m-1)!} \cdot v_1 + \frac{x^{m-2}}{(m-2)!} \cdot v_2 + \dots + x_m\right]$

=> these solutions are linearly independent

Prove that Y_k in (3) is a solution of (1') $Y_k' = AY_k$

Calculate both member

$$Y_k' = \left[\frac{x^{k-2}}{(k-2)!}\cdot v_1 + \dots + v_{k-1}\right]e^{\lambda x} + \left[\frac{x^{k-1}}{(k-1)!}\cdot v_1 + \dots + v_k\right]\cdot \lambda e^{\lambda x}$$

$$AY_k = \left[\frac{x^{k-1}}{(k-1)!}\cdot Av_1 + \dots + Av_k\right]e^{\lambda x}$$

Fundamental matrix consider IVP (initial value problem with a time variable matrix $A(x)$)

$$\begin{cases} Y'(x) = A(x)Y(x) & (1') \text{ homogeneous SDE} \\ Y(x_0) = Y_0 & (1c) \\ & \text{(initial condition)} \end{cases}$$

Determine the solutions $Y_1(x), Y_2(x), \dots, Y_n(x)$ of n IVPs

$$\begin{cases} Y_1'(x) = A(x)\cdot Y_1(x) \\ Y(x_0) = e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \end{cases}$$

$$\begin{cases} Y_2'(x) = A(x)\cdot Y_2(x) \\ Y_2(x_0) = e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \end{cases}$$

$$\dots \quad \dots \quad \dots$$

$$\begin{cases} Y_n'(x) = A(x)\cdot Y_n(x) \\ Y_n(x_0) = e_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \end{cases}$$

Remark: Actually the solutions depends on the initial moment x_0

Definition: The matrix formed with the columns $Y_1(x), \dots, Y_n$ is called fundamental of the SDE(1), the matrix $A(x)$ and it is denoted by $\Phi(x, x_0)$.

$$\Phi(x, x_0) = (Y_1(x), Y_2(x) \dots Y_n(x))$$

Properties:

1. $\frac{d}{dx}\Phi(x, x_0) = A(x) \cdot \Phi(x, x_0)$
2. $\Phi(x_0, x_0) = I_n$
3. The solution of IVP(1)(1c) is $Y(x) = \Phi(x, x_0) Y_0$ (2)
4. $\Phi(t, t_1) \cdot \Phi(t_1, t_0) = \Phi(t, t_0)$ (the subgroup property)
5. $\Phi(t, t_0)^{-1} = \Phi(t_0, t)$

Proof:

1. One derives a matrix by deriving all its elements, hence by deriving all its columns

$$\frac{d}{dx}\Phi(x, x_0) = (Y_1'(x) \quad Y_2'(x) \quad Y_n'(x)) =$$

$$= A(x)(Y_1(x) \quad Y_2(x) \quad Y_n(x))$$

2. Replace x by x_0

$\phi(x_0, x_0) = (Y_1(x_0) \quad Y_2(x_0) \quad \dots \quad Y_n(x_0)) = I_n$

3. Show that $\tilde{Y}(x) = \phi(x, x_0) \cdot Y_0$ verifies (1') and (1C)

Calculate $\tilde{Y}'(x) = \frac{d}{dx}(\phi(x, x_0) Y_0) = (A(x) Y(x, x_0) Y_0$

$A(x)(Y(x, x_0) \cdot Y_0) = A(x) \cdot Y(x)$

For $x = x_0$, $Y(x_0) = \phi(x_0, x_0) Y_0 \equiv I_n \cdot Y_0 = Y_0 \Rightarrow \tilde{Y}$ is the solution of IVP

Remark: Similarly, we can prove that an $n \times n$ matrix $M(x, x_0)$ has the properties 1 and 2 $\begin{cases} \frac{d}{dx} M(x, x_0) = A(x) M(x, x_0) \\ M(x_0, x_0) = I_n \end{cases}$

then $Y(x) = M(x, x_0) Y_0$

SEMINAR

Systems of differential equations, with constant coeff. nonhomogeneous

$$\left.\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 + b_1(x) \\ y_2' &= a_{21}y_1 + a_{22}y_2 + b_2(x) \end{aligned}\right\} + \left.\begin{aligned} y_1(0) &= \alpha \\ y_2(0) &= \beta \end{aligned}\right\} \text{initial value}$$

$$Y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix};\ A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix};\ b(x) = \begin{pmatrix} b_1(x) \\ b_2(x) \end{pmatrix};\ Y_0 = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow$$

$$\Rightarrow Y' = AY + b(x),\ Y(0) = Y_0$$

Algorithm:

1. Determine the eigenvalues of A ($\det(A - \lambda I_n) = 0$)
2. Find e^{Ax}

Case I ($\lambda_1 \neq \lambda_2$)

a. Find v_1 eigenvector for λ_1
   v_2 eigenvector for λ_2

b. Form $C = (v_1, v_2)$. Find C^{-1}

c. $e^{Dx} = \begin{pmatrix} e^{\lambda_1 x} & 0 \\ 0 & e^{\lambda_2 x} \end{pmatrix}$ and $e^{Ax} = C \cdot e^{Dx} \cdot C^{-1}$

Case II (λ, $m_a = 2$)

a. Find v eigenvector of λ and create v_{p_1}

b. $C = (v \; v_{p_1})$. Δt . C^{-1}

c. $e^{Jx} = e^{\lambda x} \begin{pmatrix} x & 1 \\ 0 & 1 \end{pmatrix}$ and $e^{Ax} = C \cdot e^{Jx} \cdot C^{-1}$

3. Calculate $y_h = e^{Ax} \cdot y_0$

4. Find $y_p = \int_0^x e^{A(x-t)} \cdot b(t)\, dt$

5. $y = y_h + y_p$

Ex.

1. $\begin{cases} y_1' = -2y_1 + 2y_2 \\ y_2' = y_1 - y_2 + e^{-3x} \end{cases}$ $\quad y_1(0) = 1$, $y_2(0) = -1$

$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$; $A = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$; $b(x) = \begin{pmatrix} 0 \\ e^{-3x} \end{pmatrix}$; $y_0 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\det(A - \lambda I_2) = \begin{vmatrix} -2-\lambda & 2 \\ 1 & -1-\lambda \end{vmatrix} = 0 \Rightarrow (-2-\lambda)(-1-\lambda) - 2 = 0$

$(2+\lambda)(1+\lambda) - 2 = 0$

$\lambda^2 + 3\lambda = 0 \Rightarrow \lambda(\lambda+3) = 0$

$\Rightarrow \lambda_1 = 0$

$\lambda_2 = -3$

$\lambda_1 = 0 \Rightarrow (A - \lambda_1 I_2) \cdot V = 0 \Rightarrow \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\Rightarrow \begin{cases} -2a + 2b = 0 \\ a - b = 0 \Rightarrow a = b \end{cases}$

$V_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\lambda_2 = 3 \Rightarrow \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow a = -2b$

$V_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$C = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$

$\det C = 1 + 2 = 3$

$C^T = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}, \quad C^* = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$

$C^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$

$e^{\Delta x} = \begin{pmatrix} e^{0 \cdot x} & 0 \\ 0 & e^{-3x} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{-3x} \end{pmatrix}$

$e^{Ax} = C \cdot e^{\Delta x} \cdot C^{-1} = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & e^{-3x} \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$

$= \frac{1}{3} \begin{pmatrix} 1 + 2e^{-3x} & 2 - 2e^{-3x} \\ 1 - 2e^{-3x} & 2 + e^{-3x} \end{pmatrix}$

$$y_{l.h.} = e^{Ax} \cdot y_0 = \frac{1}{3}\begin{pmatrix} 1+2e^{-3x} & 2-2e^{-3x} \\ 1-e^{-3x} & 2+e^{-3x} \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -1+4e^{-3x} \\ -1-2e^{-3x} \end{pmatrix}$$

$$y_p = \int_0^x e^{A(x-t)} b(t)\,dt = \int_0^x \frac{1}{3}\begin{pmatrix} 1+2e^{-3(x-t)} & 2-2e^{-3(x-t)} \\ 1-e^{-3(x-t)} & 2+e^{-3(x-t)} \end{pmatrix}\begin{pmatrix} 0 \\ e^{-3x} \end{pmatrix}$$

$$= \frac{1}{3}\int_0^x \begin{pmatrix} 2e^{-3t} - 2e^{3x} \\ 2e^{-3t} + e^{-3x} \end{pmatrix} dt = \frac{1}{3}\begin{pmatrix} I_1 = \int_0^x (2e^{-3t} + 2e^{-3x})\,dt \\ I_2 = \int_0^x (2e^{-3t} + e^{-3x})\,dt \end{pmatrix}$$

$$I_1 = \frac{2e^{-3t}}{-3}\Big|_0^x - 2e^{-3x}t\Big|_0^x = -\frac{2}{3}(e^{-3x} - 1) - 2xe^{-3x}$$

$$I_2 = \frac{2e^{-3t}}{-3}\Big|_0^x + e^{-3x}\Big|_0^x = -\frac{2}{3}(e^{-3x} - 1) + xe^{-3x}$$

$$y = y_{l.h.} + y_p = \frac{1}{3}\begin{pmatrix} -\frac{1}{3} + \frac{10}{3}e^{-3x} - 2xe^{-3x} \\ -\frac{1}{3} - \frac{4}{3}e^{-3x} + xe^{-3x} \end{pmatrix}$$

2. $$\begin{cases} y_1' = 2y_2 + 1 & y_1(0) = 0 \\ y_2' = -2y_1 + 4y_2 + x & y_2(0) = 2 \end{cases}$$

$$A = \begin{pmatrix} 0 & 2 \\ -2 & 4 \end{pmatrix}$$

$$\det(A - \lambda i_2) = 0 \Rightarrow \begin{vmatrix} -\lambda & 2 \\ -2 & 4-\lambda \end{vmatrix} = -4\lambda + \lambda^2 + 4 = 0$$

$$(\lambda - 2)^2 = 0 \Rightarrow \lambda = 2$$

$$(A - 2i_2)\cdot V = 0 \Rightarrow \begin{pmatrix} -2 & 2 \\ -2 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} -2x + 2y = 0 \\ -2x + 2y = 0 \end{cases} \Rightarrow x = y$$

$$V_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$(A-\lambda I_2)\cdot V_p = V \Rightarrow \begin{pmatrix} -2 & 2 \\ -2 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$-2x+2y=1 \Rightarrow y = \frac{1+2x}{2}$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ \frac{1}{2}+x \end{pmatrix} = \begin{pmatrix} x \\ x \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix} \rightarrow V_p$

$C = \begin{pmatrix} 1 & 0 \\ 1 & \frac{1}{2} \end{pmatrix} = 1$

$\det C = \frac{1}{2}$; $C^T = \begin{pmatrix} 1 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$; $C^* = \begin{pmatrix} \frac{1}{2} & -1 \\ 0 & 1 \end{pmatrix}$

$C^* = \begin{pmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{pmatrix}$

$C^{-1} = 2\begin{pmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{pmatrix}$

$e^{Jx} = e^{2x}\begin{pmatrix} x & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} xe^{2x} & e^{2x} \\ 0 & e^{2x} \end{pmatrix}$

$e^{Ax} = C\cdot e^{Jx}\cdot C^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & \frac{1}{2} \end{pmatrix}\begin{pmatrix} xe^{2x} & e^{2x} \\ 0 & e^{2x} \end{pmatrix}\begin{pmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{pmatrix}$

$= 2\cdot\begin{pmatrix} xe^{2x} & e^{2x} \\ xe^{2x} & e^{2x}+\frac{e^{2x}}{2} \end{pmatrix}\begin{pmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{pmatrix} = 2\begin{pmatrix} \frac{xe^{2x}}{2}-e^{2x} & e^{2x} \\ \frac{xe^{2x}}{2}-e^{2x}-\frac{e^{2x}}{2} & e^{2x}+\frac{e^{2x}}{2} \end{pmatrix}$

$= \begin{pmatrix} xe^{2x}-2e^{2x} & 2e^{2x} \\ xe^{2x}-3e^{2x} & 3e^{2x} \end{pmatrix}$

$Y_h = \begin{pmatrix} \frac{xe^{2x}}{2}-e^{2x} & e^{2x} \\ \frac{xe^{2x}}{2}-e^{2x}-\frac{e^{2x}}{2} & e^{2x}+ \end{pmatrix}$

$$Y_h = \begin{pmatrix} xe^{2x} - 2e^{2x} & 2e^{2x} \\ xe^{2x} - 3e^{2x} & 3e^{2x} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 4e^{2x} \\ 6e^{2x} \end{pmatrix}$$

$$Y_p = \int_0^x \begin{pmatrix} 4e^{2x-2t} \\ 6e^{2x-2t} \end{pmatrix} \begin{pmatrix} 1 \\ t \end{pmatrix} dt = \int_0^x \begin{pmatrix} 4e^{2x-2t} + 4te^{2x-2t} \\ 6e^{2x-2t} + 6te^{2x-2t} \end{pmatrix} dt$$

$$= I_1 =$$

$$Y_p = \int_0^x \begin{pmatrix} xe^{2x-2t} - 2e^{2x-2t} + 2te^{2x-2t} \\ xe^{2x-2t} - 3e^{2x-2t} + 3te^{2x-2t} \end{pmatrix} dt$$

$$I_1 = \int_0^x xe^{2x-2t} - 2e^{2x-2t} + 2te^{2x-2t} dt = \int_0^x \frac{xe^{2x}}{e^{2t}} - \frac{2e^{2x}}{e^{2t}} + \frac{2te^{2x}}{e^{2t}} dt$$

$$= e^{2x} \int_0^x \frac{x - 2 + 2t}{e^{2t}} dt = e^{2x} \left(\frac{e^{-2t}}{-2} \cdot x + \frac{e^{-2t}}{2} \right.$$

$$\int \frac{2t}{e^{2t}} = 2t$$

$$I_1 = \int_0^x e^{2x-2t}(x + t - 2)\, dt = \int_0^x \left(\frac{e^{2x-2t}}{-2} \right)' (x + t - 2)\, dt = \frac{e^{2x-2t}}{-2}(x + t - 2)$$

$$- \int_0^x \frac{e^{2x-2t}}{-2} dt = (-x - 1) + \frac{e^{2x}}{2}(x - 2) + \frac{1}{2} \frac{e^{2x-2t}}{-2} \Big|_0^x = -x + 1 + \frac{e^{2x}}{2}(x - 2) - \frac{1}{4}$$

$$+ \frac{e^{2x}}{4} = -x + \frac{3}{4} + e^{2x}\left(\frac{x}{2} - \frac{3}{4} \right)$$

$$I_2 = (-x - 1) \frac{e^{2x}}{2}(x - 2) + \frac{3}{2} \frac{e^{2x-2t}}{-2} \Big|_0^x = (-x - 1) + \frac{e^{2x}}{2}(x - 2) - \frac{1}{2}$$

$$Y = \begin{pmatrix} 4e^{2x} \\ 6e^{2x} \end{pmatrix} \begin{pmatrix} -x + \frac{3}{4} + 2t^2 e^{2x}\left(\frac{x}{2} - \frac{3}{4} \right) \\ -(-x + 1) + \frac{e^{2x}}{2}(x - 2) - \frac{1}{2} \end{pmatrix}$$

$$I_2 = \frac{e^{2x-2t}}{-2}(x + 2t - 3) \Big|_0^x - \int_0^x \frac{e^{2x-2t}}{-2} \cdot 2\, dt =$$

$$= -(x - 3) + -\frac{1}{2}(3x - 3) + \frac{e^{2x}}{2}(x - 3) - \frac{e^{2x-2t}}{-2} \Big|_0^x = -\frac{1}{2}(3x - 3) + \frac{e^{2x}}{2}(x - 3)$$

$$+ \frac{1}{2} - \frac{e^{2x}}{2}$$

$$I_2 = -\frac{3x}{2} + \frac{x}{2} + \frac{e^{2x}\cdot x}{2} - \frac{3e^{2x}}{2} + \frac{1}{2} - \frac{e^{2x}}{2} =$$

$$= -x - 2e^{2x} + \frac{e^{2x}x}{2} + \frac{1}{2}$$

$$y = y_h + y_p$$

3.
$$\begin{cases} y_1' = 3y_1 + 4y_2 + e^{2x} & y_1(0) = 1 \\ y_2' = y_1 + 3y_2 + e^{x} & y_2(0) = 1 \end{cases}$$

$$A = \begin{pmatrix} 3 & 4 \\ 1 & 3 \end{pmatrix}$$

$$\det(A - \lambda I_2) = 0 \Rightarrow \begin{vmatrix} 3-\lambda & 4 \\ 1 & 3-\lambda \end{vmatrix} = 0 \Rightarrow 9 - 6\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 6\lambda + 5 = 0$$

$$\Delta = 36 - 20 = 16$$

$$\lambda_{1,2} = \frac{6 \pm 4}{2} \begin{cases} 1 \\ 5 \end{cases}$$

$$\lambda_1 = 1 \Rightarrow (A - \lambda_1 I_2)\cdot V = 0 \Rightarrow \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} 2a + 4b \\ a + 2b = 0 \Rightarrow a = -2b \end{cases}$$

$$\begin{pmatrix} -2 \\ 1 \end{pmatrix} = V_1$$

$$\lambda_2 = 5 \Rightarrow (A - \lambda_1 I_2)\cdot V = 0 \Rightarrow \begin{pmatrix} -2 & 4 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$a = 2k$

$v_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$C = \begin{pmatrix} -2 & 2 \\ 1 & 1 \end{pmatrix}$

$\det C = -2 - 2 = -4$

$C^T = \begin{pmatrix} -2 & 1 \\ 2 & 1 \end{pmatrix}$

$C^* = \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix}$

$C^{-1} = \frac{1}{-4} \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix}$

$e^{\Delta x} = \begin{pmatrix} e^x & 0 \\ 0 & e^{5x} \end{pmatrix}$

$e^{Ax} = C \cdot e^{\Delta x} \cdot C^{-1} = -\frac{1}{4} \begin{pmatrix} -2 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} e^x & 0 \\ 0 & e^{5x} \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix}$

$= -\frac{1}{4} \begin{pmatrix} -2e^x & 2e^{5x} \\ e^x & e^{5x} \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix} = -\frac{1}{4} \begin{pmatrix} -2e^x - 2e^{5x} & 4e^x + 4e^{5x} \\ e^x - e^{5x} & -2e^x + 2e^{5x} \end{pmatrix}$

$y_k = e^{Ax} \cdot y_0 = -\frac{1}{4} \begin{pmatrix} -2e^x - 2e^{5x} & 4e^x + 4e^{5x} \\ e^x - e^{5x} & -2e^x + 2e^{5x} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 2e^x + 2e^{5x} \\ -e^x + e^{5x} \end{pmatrix}$

COURSE

Fundamental matrix. Initial value problem (IVP)

$$\begin{cases} Y'(x) = A(x)Y(x) \quad (1') \text{ homogeneous SDE} \\ Y(x_0) = Y_0 \end{cases}$$

We determined n solutions $Y_1(x), Y_2(x), \dots, Y_n(x)$ corresponding $x_0 = l_1 + l_2 + \dots + l_n$

Fund $\Rightarrow$ Unfundamental matrix $\Phi(x, x_0) = (Y_1(x), Y_2(x) \dots Y_n(x))$

Proprieties:

1. $\frac{d}{dx}\Phi(x, x_0) = A(x)\Phi(x, x_0)$
2. $\Phi(x_0, x_0) = I_n$
3. The solution of the IVP $(1') \Rightarrow \underline{Y}(x) = \Phi(x, x_0)Y_0$

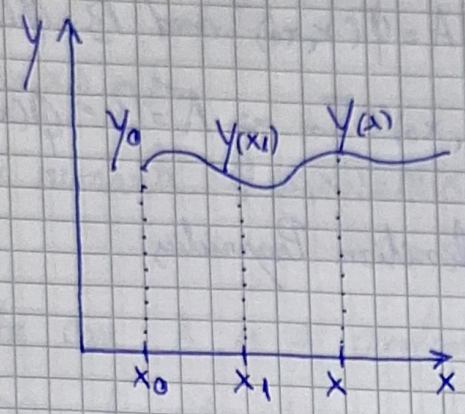
Remark: If a matrix $M(x, x_0)$ has the properties 1), 2) i.e. $\frac{d}{dx}M(x, x_0) = A(x)M(x, x_0)$, $M(x_0, x_0) = I_n$ then

$$\Phi(x, x_0) = M(x, x_0)$$

4. $\Phi(x, x_1)\Phi(x_1, x_0) = \Phi(x, x_0)$
5. $\Phi(x, x_0)^{-1} = \Phi(x_0, x)$

Proof:

4. Apply property 3) in interval $[x_0, x_1]$, $[x_1, x]$, $[x_0, x]$



On $[x_0, x_1]$ $y(x_1) = \phi(x_1, x_0) y_0$

On $[x_1, x]$ $y(x) = \phi(x, x_1) y(x_1)$ } Replace $y(x_1)$

On $[x_0, x]$ $y(x) = \phi(x, x_0) y_0$

We obtain the solution of the IVP $\Rightarrow y(x) = \phi(x, x_1)\phi(x_1, x_0) y_0$

$\Rightarrow$ two solutions

But the IVP has a unique solution $\Rightarrow \phi(x, x_1)\phi(x_1, x_0) y_0 = \phi(x, x_0) y_0$

$\Rightarrow$ the matrices are equal $\Rightarrow \phi(x, x_1)\phi(x_1, x_0) = \phi(x, x_0)$

5. Let A and B be two $n \times n$ matrices

Then B is the inverse of the matrix A i.e $A^{-1} = B \Leftrightarrow B \cdot A = I_n$

Take $A = \phi(x, x_0)$ and $B = \phi(x_0, x) \Rightarrow$

$BA = \phi(x_0, x)\phi(x, x_0) = \phi(x_0, x_0) = I_n \Rightarrow A^{-1} = B \Rightarrow \phi(x, x_0)^{-1} = \phi(x_0, x)$

Ex: Fundamental matrix. Definition. Properties

Properties:

6. Peano - Baker formula

$$\phi(x, x_0) = I_n + \int_{x_0}^{x} A(s_1)\, ds_1 + \int_{x_0}^{x} A(s_1) \int_{x_0}^{s_1} A(s_2)\, ds_2 ds_1 + \dots +$$

$$+ \int_{x_0}^{x} A(s_1) \int_{x_0}^{s_1} A(s_2) \dots \int_{x_0}^{s_{k-1}} A(s_k)\, ds_k\, ds_{k-1} \dots ds_1$$

Denote with $M(x, x_0)$ the sum of the series of ~~mod~~

Prove that $M(x, x_0)$ has the properties (1), (2) By remark $\Rightarrow \phi(x, x_0) = M(x, x_0)$

Idea: Derive term by term, the series

$$\frac{d}{dx} M(x, x_0) = 0_n + A(x) + A(x) \int_{x_0}^{x} A(s_2)\, ds_2 + \dots + A(x) \int_{x_0}^{x} A(s_2) \dots$$

$$= A(x) \left[I_n + \int_{x_0}^{x} A(s_2)\, ds_2 + \dots + \int_{x_0}^{x} A(s_2) \dots \int_{x_0}^{s_{k-1}} A(s_k)\, ds_k \dots ds_2 \right]$$

$\frac{d}{dx} M(x, x_0) = A(x) M(x, x_0)$

↳ replace x by x_0

$M(x_0, x_0) = I_n$

By remark $\Phi(x, x_0) = M(x, x_0)$ hence (PB) is true

7. The case: A is a constant $n \times n$ matrix

(PB) formula begins:

$$\Phi(x, x_0) = I_n + A\int_{x_0}^{x} ds_1 + A^2 \int_{x_0}^{x}\left(\int_{x_0}^{s_1} ds_2\right) ds_1 + \ldots + A^k \int_{x_0}^{x}\left(\int_{x_0}^{s_1} \ldots \int_{x_0}^{s_{k-1}} ds_k \ldots ds_1\right)$$

Calculate the integrals $\int dx = x \Rightarrow \int_{x_0}^{x} ds_1 = s_1\Big|_{x_0}^{x} = x - x_0 = \frac{x - x_0}{1!}$

$$\int_{x_0}^{x}\left(\int_{x_0}^{s_1} ds_2\right) ds_1 = \int_{x_0}^{x} \frac{s_1 - x_0}{1!} ds_1 = \frac{(s_1 - x_0)^2}{2 \cdot 1!}\Big|_{x_0}^{x} = \frac{(x - x_0)^2}{2!}$$

Assume that this is true for $k-1 \Rightarrow$

$\Rightarrow k-1$ integrals $= \frac{(\text{difference of limits})^{k-1}}{(k-1)!}$

Prove for k integrals

$$\int_{x_0}^{x}\left(\int_{x_0}^{s_1} \ldots \int_{x_0}^{s_{k-1}} ds_k \ldots ds_2\right) ds_1 = \int_{x_0}^{x} \frac{(s_1 - x_0)^{k-1}}{(k-1)!} ds_1 = \frac{(s_1 - x_0)^k}{k(k-1)!} = \frac{(x - x_0)^k}{k!}$$

The series of Φ becomes

$$\Phi(x, x_0) = I_n + \frac{A(x - x_0)}{1!} + \frac{(A(x - x_0))^2}{2!} + \ldots + \frac{(A(x - x_0))^k}{k!}$$

$e^{A(x - x_0)}$ (exponential matrix)

5. Let A and B be two $n \times n$ matrices

Then B is the inverse of the matrix A i.e $A^{-1} = B \Leftrightarrow B \cdot A = I_n$

Take ~~A=(x,x0) and B(x~~ $A = \phi(x, x_0)$ and $B = \phi(x_0, x) \Rightarrow$

$BA = \phi(x_0, x)\phi(x, x_0) = \phi(x_0, x_0) = I_n \Rightarrow A^{-1} = B \Rightarrow \phi(x, x_0)^{-1} = \phi(x_0, x)$

Ex: Fundamental matrix. Definition. Properties

Properties:

6. Peano - Baker formula

$$\phi(x, x_0) = I_n + \int_{x_0}^{x} A(s_1)\, ds_1 + \int_{x_0}^{x} A(s_1) \int_{x_0}^{s_1} A(s_2)\, ds_2 ds_1 + \dots +$$

$$+ \int_{x_0}^{x} A(s_1) \int_{x_0}^{s_1} A(s_2) \dots \int_{x_0}^{s_{k-1}} A(s_k)\, ds_k ds_{k-1} \dots ds_1$$

Denote with $M(x, x_0)$ the sum of the series of ~~mat~~

Prove that $M(x, x_0)$ has the properties (1), (2) Key remark $\Rightarrow \phi(x, x_0) = M(x, x_0)$

Idea: Derive term by term, the series

$$\frac{d}{dx} M(x, x_0) = 0_n + A(x) + A(x) \int_{x_0}^{x} A(s_2)\, ds_2 + \dots + A(x) \int_{x_0}^{x} A(s_2) \dots$$

$$= A(x) \left[I_n + \int_{x_0}^{x} A(s_2)\, ds_2 + \dots + \int_{x_0}^{x} A(s_2) \dots \int_{x_0}^{s_{k-1}} A(s_k)\, ds_k \dots ds_2 \right]$$

$\frac{d}{dx} M(x,x_0) = A(x) M(x,x_0)$

$\Leftrightarrow$ replace x by x_0

$M(x_0,x_0) = I_n$

By remark $\Phi(x,x_0) = M(x,x_0)$ hence (PB) is true

7. The case: A is a constant $n \times n$ matrix

(PB) formula begins:

$$\Phi(x,x_0) = I_n + A\int_{x_0}^{x} ds_1 + A^2 \int_{x_0}^{x}\left(\int_{x_0}^{s_1} ds_2\right) ds_1 + \ldots + A^k \int_{x_0}^{x}\left(\int_{x_0}^{s_1} \ldots \int_{x_0}^{s_{k-1}} ds_k \ldots ds_1\right)$$

Calculate the integrals $\int dx = x \Rightarrow \int_{x_0}^{x} ds_1 = s_1\Big|_{x_0}^{x} = x - x_0 = \frac{x-x_0}{1!}$

$$\int_{x_0}^{x}\left(\int_{x_0}^{s_1} ds_2\right) ds_1 = \int_{x_0}^{x} \frac{s_1 - x_0}{1!}\, ds_1 = \frac{(s_1-x_0)^2}{2\cdot 1!}\Big|_{x_0}^{x} = \frac{(x-x_0)^2}{2!}$$

Assume that this is true for $k-1 \Rightarrow$

$\Rightarrow k-1$ integrals $= \frac{(\text{difference of limits})^{k-1}}{(k-1)!}$

Prove for k integrals

$$\int_{x_0}^{x}\left(\int_{x_0}^{s_1} \ldots \int_{x_0}^{s_{k-1}} ds_k \ldots ds_2\right) ds_1 = \int_{x_0}^{x} \frac{(s_1-x_0)^{k-1}}{(k-1)!}\, ds_1 = \frac{(s_1-x_0)^k}{k(k-1)!} = \frac{(x-x_0)^k}{k!}$$

The series of Φ becomes

$$\Phi(x,x_0) = I_n + \frac{A(x-x_0)}{1!} + \frac{(A(x-x_0))^2}{2!} + \ldots + \frac{(A(x-x_0))^k}{k!}$$

$= e^{A(x-x_0)}$ (exponential matrix)

3) becomes (for $x_0 = 0$) $\Rightarrow y(x) = e^{Ax} y_0$

8. The solution of the IVP* (1) (1c)

IVP $\begin{cases} y'(x) = A(x)y(x) + b(x) & (1) \text{ nonhomogeneous SDE} \\ y(x_0) = y_0 & (1c) \end{cases}$

is given by the variation of parameter formula

$$y(x) = \Phi(x, x_0) y_0 + \int_{x_0}^{x} \Phi(x, s) b(s)\, ds \quad (VPF)$$

Obs: 8) Without proof at the exam

9. For A constant matrix i.e $\Phi(x, x_0) = e^{A(x-x_0)}$ (VPF) becomes

$$y(x) = e^{A(x-x_0)} y_0 + \int_{x_0}^{x} \left(s^{A(x-s)} \right) b(s)\, ds$$

CHAPTER II STABILITY

1. Stability of linear, homogenious SDEs with constant coefficients

IVP: $\begin{cases} Y'(x) = A\,Y(x) & (1') \\ Y(x_0) = Y_0 & (1C) \end{cases}$

The solution is $Y(x) = e^{A(x-x_0)}\,Y_0$

Take $x_0 = 0 \Rightarrow Y(x) = e^{Ax} \cdot Y_0$ (1')

For $Y_0 = \bar{0} \Rightarrow Y(x) = e^{Ax} \cdot \bar{0} = \bar{0}$

if the initial position is null, then the solution of the SDE contain the null potential

SEMINAR

Stability

$Y' = AY$ - only asimptotically stable system

Method 1:

1. Find the eigenvalues of A
2. If, for all eigenvalues λ, you have $Re(\lambda) < 0 \Rightarrow$
$\Rightarrow$ asimptotically stable system

Method 2 (Routh-Hurwitz Criterion)

1. Calculate $P_A(\lambda) = \det(\lambda I_n - A) = -a_0\lambda^{2n} + a_1\lambda^{n-1} + \dots + a_n$
2. Check that $a_j > 0$, $j = \overline{0, n}$ and create the Hurwitz matrix

$$H_n = \begin{pmatrix} a_1 & a_0 & 0 & \dots & 0 \\ a_3 & a_2 & a_1 & a_0 & 0 \dots 0 \\ \dots & \dots & \dots & \dots & \dots \\ a_{2n-1} & a_{2n-2} & \dots & \dots & a_0 \end{pmatrix}, \quad a_k = 0, \; k > n$$

If H_n is a positive defined matrix $\Rightarrow$ the system is ~~asym~~ asimptotically stable

Method 3 (Lyapunov)

If, for a certain matrix Q, symmetric and positive defined, we can find a matrix P, also symmetric and positive defined, so that

$A^T \cdot P + P \cdot A = -Q$

then the system is as. stable

Check using all the 3 methods

1. $A = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$; $Q = \begin{pmatrix} 8 & 6 \\ 6 & 4 \end{pmatrix}$

Method 1:

$$\det(A - \lambda I_2) = 0 \Rightarrow \begin{vmatrix} -1-\lambda & 1 \\ -1 & -1-\lambda \end{vmatrix} = (-1-\lambda)^2 + 1 = 0 \Rightarrow \lambda^2 + 2\lambda + 2 = 0$$

$$\Delta = -4$$

$$\lambda_{1,2} = \frac{-2 \pm 2i}{2} \begin{cases} -1+i \\ -1-i \end{cases}$$

$Re(-1+i) = -1 < 0$

$Re(-1-i) = -1 < 0$

$\Rightarrow$ asymptotically stable

Method 2

$P_A(\lambda) = \det(\lambda I_2 - A) = \lambda^2 + 2\lambda + 2$

$a_0 = 1$

$a_1 = 2$

$a_2 = 2$

$H_2 = \begin{pmatrix} a_1 & a_0 \\ a_3 & a_2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ $\quad \Delta_1 = 2 > 0$, $\Delta_2 = 4 > 0$ $\Rightarrow$ positive defined

$\Rightarrow$ as. stable system

Method 3

$P = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$

$A^T P + P A = -Q \Rightarrow \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} a & b \\ b & c \end{pmatrix} + \begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 8 & 6 \\ 6 & 4 \end{pmatrix}$

$\begin{cases} -2a - 2b = -8 \mid :-2 \\ a - 2b - c = -6 \\ +a - 2b - c = -6 \\ 2b - 2c = -4 \end{cases}$ $\Rightarrow a + b = 4 \Rightarrow b = 4 - a \Rightarrow b = 2$

$b - c = -2 \Rightarrow c = b + 2 = 6 \div a \Rightarrow c = 4$

$a - 8 + 2a - 6 + a = -6 \Rightarrow 4a - 14 = -6$

$4a = 8 \Rightarrow a = 2$

$P = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$ $\begin{matrix} \Delta_1 = 2 > 0 \\ \Delta_2 = 4 > 0 \end{matrix}$ $\Big\} \Rightarrow$ as. stable system

2. $A = \begin{pmatrix} 0 & 0 & -2 \\ -1 & -1 & 1 \\ 1 & 0 & -3 \end{pmatrix}$; $Q = \begin{pmatrix} 4 & 1 & 4 \\ 1 & 2 & 3 \\ 4 & 3 & 24 \end{pmatrix}$

I. $P(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} -\lambda & 0 & -2 \\ -1 & -1-\lambda & 1 \\ 1 & 0 & -3-\lambda \end{vmatrix} = 0$

$(-1-\lambda)(-1)^{2+2} \begin{vmatrix} -\lambda & -2 \\ 1 & -3-\lambda \end{vmatrix} = (-1-\lambda)(3\lambda + \lambda^2 + 2)$

$\Rightarrow \lambda_1 = -1$

$\lambda^2 + 3\lambda + 2 = 0 \Rightarrow$

$\Delta = 9 - 8 = 1$

$\lambda_{1,2} = \frac{-3 \pm 1}{2} \begin{matrix} \nearrow -2 \\ \searrow -1 \end{matrix}$

$Re(-1) < 0$

$Re(-2) < 0$

$Re(-1) < 0$

II. $P_A(\lambda) = \det(\lambda I_3 - A) = (-1)^3(-1-\lambda)(3\lambda+\lambda^2+2)$

$= (1+\lambda)(3\lambda+\lambda^2+2)$

$3\lambda+\lambda^2+2+3\lambda^2+\lambda^3+2\lambda$

$\lambda^3+4\lambda^2+5\lambda+2=0$

$a_0=1 \quad a_2=5$

$a_1=4 \quad a_3=2$

$$H_3 = \begin{pmatrix} a_1 & a_0 & 0 \\ a_3 & a_2 & a_1 \\ a_5 & a_4 & a_3 \end{pmatrix} = \begin{pmatrix} 4 & 1 & 0 \\ 2 & 5 & 4 \\ 0 & 0 & 2 \end{pmatrix} =$$

$\Delta_1 > 0 \qquad \Delta_1 = 4$

$\Delta_2 > 0 \qquad \Delta_2 = 18$

$\Delta_3 > 0 \qquad \Delta_3 = 2 \cdot \Delta_2 = 36$.

III. $A^T = \begin{pmatrix} 0 & -1 & -2 \\ 0 & -1 & 1 \\ -2 & 1 & -3 \end{pmatrix}$

$P = \begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix}$

$A^T \cdot P + P \cdot A = -Q \Rightarrow \begin{pmatrix} -a-2d & -b-2f & -f-2c \\ -d+e & -b+f & -f+c \\ -2a+d-3e & -2d+b-3f & -3 \end{pmatrix}$

$-2b+2c=-4$

$-e-b+d=-1$

$-2a+b-3c-d+f=-4$

$-b-e+d=-1$

$-2e=-2$

3. $A = \begin{pmatrix} -3 & 1 & 0 \\ -1 & -3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$; $C_2 = \begin{pmatrix} 8 & 7 & 0 \\ 7 & 10 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

I. $P(\lambda) = \det(A - \lambda i_3) = \begin{vmatrix} -3-\lambda & 1 & 0 \\ -1 & -3-\lambda & 0 \\ 0 & 0 & -1-\lambda \end{vmatrix} = 0$

$(-1-\lambda)((-3-\lambda)^2 + 1) = 0$

$(-1-\lambda)(9 + 6\lambda + \lambda^2 + 1) = 0$

$\lambda = -1$

$\lambda^2 + 6\lambda + 10 = 0$

$\Delta = 64 - 40 = 2$ ~~36 - 40 = -4~~

$\lambda_{2,3} = \frac{-6 \pm 2i}{2}$ $\begin{cases} -3 + 1i \\ -3 - 1i \end{cases}$

$Re(-1) < 0$

$Re(-3+2i) < 0$ $\Rightarrow$ as. stable sys.

$Re(-3-2i) < 0$

II. $P_A(\lambda) = \det(\lambda i_3 - A) = (-1)^3 \det(A - \lambda i_3) = 0$

$= (1+\lambda)(\lambda^2 + 6\lambda + 10) = 0$

$= \lambda^2 + 6\lambda + 10 + \lambda^3 + 6\lambda^2 + 10\lambda$

$\lambda^3 + 7\lambda^2 + 16\lambda + 10$

$a_0 = 1$; $a_1 = 7$

$a_2 = 16$; $a_3 = 10$

$$M_3 = \begin{pmatrix} 7 & 1 & 0 \\ 10 & 16 & 7 \\ 0 & 0 & 10 \end{pmatrix}$$

$\Delta_1 > 0$

$\Delta_2 > 0$

$\Delta_3 > 0$

III. $A^T = \begin{pmatrix} -3 & -1 & 0 \\ 1 & -3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$; $A = \begin{pmatrix} -3 & 1 & 0 \\ -1 & -3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

$$P = \begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix}$$

$$A^T \cdot P + \underbrace{A \cdot P}_{PA} = -Q \Rightarrow \begin{pmatrix} -3a-d & -3d-b & -3e-f \\ a-3d & d-3b & e-3f \\ -e & -f & -c \end{pmatrix}$$

$$+ \begin{pmatrix} -3a+d & -3d+b & -3e+f \\ -a-3d & -d-3b & -e-3f \\ -e & -f & -c \end{pmatrix} = \begin{pmatrix} -8 & -7 & 0 \\ -7 & -10 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} -6a & -6d & -6e \\ -6d & -6b & -6f \\ -2e & -2f & -2c \end{pmatrix} = \begin{pmatrix} -8 & -7 & 0 \\ -7 & -10 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$-6a = -8$

$-6d = -7$

$-6e = 0$

$-6d = -7$

$-6b = -10$

$$\begin{pmatrix} -3a-d & -3d-b & -3e-f \\ -a-3d & -d-3b & -e-3f \\ -g-l & -f & -c \end{pmatrix} + \begin{pmatrix} -3a-d & a-3d & -e \end{pmatrix}$$

COURSE

Test 9 Noiembrie (BN104)

Stability of the SDEs

Homogeneous SDE with constant coefficients

$$\begin{cases} Y'(x) = AY(x) \quad (1') \\ Y(0) = Y_0 \quad (1C) \end{cases}$$

A is a $n \times n$ matrix (constant) $\Rightarrow$ The solution is $Y(x) = e^{Ax} Y_0(x)$

Remark: If the initial position is $Y_0 = \bar{0}_n$ then $Y(x) = e^{Ax} \cdot \bar{0} = \bar{0}$

$\Rightarrow Y_0 = \bar{0}$ is an equilibrium position

Definition:

1. The system (1') is stable if $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $\forall Y_0 \in \mathbb{R}^n$ with $\|Y_0\| < \delta$, the solution $Y(x) = e^{Ax} \cdot Y_0$ verifies $\|Y(x)\| < \varepsilon$

2. The system (1') is stable unstable if it is not stable

3. System (1') is asymptotically stable if it is stable and $\lim_{x \to \infty} y(x) = \bar{0}$

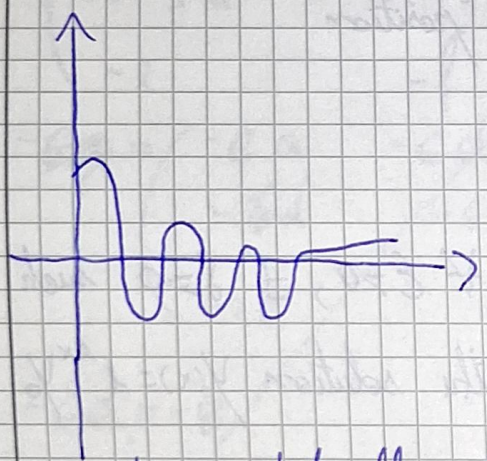
$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} ; \quad \|y\| = \sqrt{y_1^2 + y_2^2 + \dots + y_n^2}$$

Consider the case $n=1$ $y = (y) \Rightarrow \|y\| = \sqrt{y^2} = |y|$

$\|y\| < \varepsilon \Rightarrow -\varepsilon < y < \varepsilon$

Stable

Unstable



Asymptotically stable

Remark:

Let $\lambda \in \nabla(A)$ (λ is a λ eigenvalue of A)

↑ spectrum of matrix A

If $ma(\lambda) = mg(\lambda)$ ⇒ for λ, the solution $Y(x) = e^{Ax}$ Y0 can contain only simple function of the form $Ve^{\lambda x}$ in the case A is diagonalizable

↑ algebraic multitude of λ ↖ geometric multitude of λ

If $ma(\lambda) > mg(\lambda)$ not enought eigenvectors ⇒ we need in solutions chains of principal vectors (⇒ the matrix A is not diagonalizable ⇒ the solution $Y(x) =$

Proposition:

1. If $ma(\lambda) = mg(\lambda)$ $Y(x)$ (solution) and e^{Ax} can contain only functions of the form $e^{\lambda x}$
2. If $ma(\lambda) > mg(\lambda)$ $Y(x)$ and e^{Ax} can contain functions of the form $\overline{x}_{\lambda}^{\lambda x c}$

Theorem:

1. The system $y'(x) = Ay(x)$ is asymptotically stable $\Leftrightarrow$ $\forall \lambda \in \sigma(A)$, $Re(\lambda) < 0$ (all eigenvalues λ have real parts)
2. $\exists \lambda \in \sigma(A)$ with $Re(\lambda) > 0$ then the system is unstable
3. If $\forall \lambda \in Re(\lambda) \in \sigma(A)$, $Re(\lambda) = 0$ and for λ with $Re(\lambda) = 0$ $ma(\lambda) = mg(\lambda)$ then the system is stable, but not asymptotically stable
4. If $\exists \lambda \in \sigma(A)$ with $Re(\lambda) = 0$ and $ma(\lambda) > mg(\lambda)$ then the system is unstable

Proof:

A For $z \in \mathbb{C}$, $|e^z| = e^{Re\, z}$

$z \in \mathbb{C} \Rightarrow z = x + iy \qquad e^z = e^{x+iy} = e^x e^{iy}$

B. If $a>0 \Rightarrow \lim_{x\to\infty} x^\beta e^{ax} = \infty$

If $a<0$, $a=-b$, then $\lim_{x\to\infty} x^\beta e^{ax} = \lim_{x\to\infty} x^\beta e^{-bx}$ (apply L'Hopital)

$\Rightarrow \lim_{x\to\infty} x^\beta e^{ax} = 0$

$|x^\beta e^{\lambda x}| = x^\beta e^{Re(\lambda)x}$

I. equivalent statement

The system is asymptotically stable $\Leftrightarrow$ stable and $\lim y(x) = \bar{0}$ $\Leftrightarrow$ all the functions contain the solution $y(x) = e^{Ax}$ $x\to\infty$ tend to $0 \Rightarrow \forall$ function $x^\beta e^{\lambda x} \to 0$

II. If $\exists \lambda \in V(A)$ with $Re(\lambda) > 0$ then $\lim_{x\to\infty} x^\beta e^{\lambda x} = \lim_{x\to\infty} |x^\beta e^{\lambda x}| = \infty$
$\Rightarrow$ the system is unstable

III. $Re(\lambda) \leq 0$, $\forall \lambda \in V(A)$ and for λ with $Re(\lambda) = 0$ $m_a(\lambda) = m_g(\lambda)$

For λ with $Re(\lambda) < 0$ $\lim_{x\to\infty} x^\beta e^{\lambda x} = 0$

For $Re(\lambda) = 0$ can contain only simple functions $e^{\lambda x}$

then $|e^{\lambda x}| = e^{(Re\lambda)x} = e^{0x} = 1$ $\Rightarrow e^{\lambda x}$ is bounded
$\Rightarrow$ NOT 0

Conclusion: all the functions contained in the solution $y(x) = e^{Ax} y_0$ are bounded, and the same are bounded in the matrix $e^{Ax} \Rightarrow$ the matrix function $\leftrightarrow e^{Ax}$ is bounded $\Rightarrow \exists M > 0$ such that $\|e^{Ax}\| \leq M$
(majorant)

For an arbitrary $\varepsilon > 0$, take $\delta = \frac{\varepsilon}{M}$

Then, for any with Y_0 with $\|Y_0\| < 0$ i.e $\|Y_0\| \leq \frac{\varepsilon}{M}$, the solution there for verifies $\|Y(x)\| = \|e^{Ax} Y_0\| \leq M \cdot \varepsilon$

$\forall \varepsilon > 0$, $\exists \delta = \frac{\varepsilon}{M}$ such that $\forall Y_0$ with $\|Y_0\| < \delta$, the solution verifies $\|Y(x)\| < \varepsilon \iff$ the system is stable, but $|e^{\lambda x}| \not\to 0$

$\Rightarrow$ the system is not asymptotically stable

IV. If $\exists \lambda \in V(A)$ with $\mathrm{Re}(\lambda) = 0$ and $\mathrm{mal}(\lambda) > \mathrm{mg}(\lambda)$, then the solution $Y(x)$ can contain functions of the form $x^\beta e^{\lambda x}$

Then $|x^\beta e^{\lambda x}| = x^\beta e^{\mathrm{Re}\lambda \cdot x} = x^\beta e^0 = x^\beta \to \infty$

SEMINAR

1. $f: \mathbb{C} \to \mathbb{C}, \; f(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$
2. $f: \mathbb{C} \to \mathbb{C}, \; f(z) = e^z \sim e^{a+bi} = e^a (\cos b + i \sin b)$
3. $f: \mathbb{C} \to \mathbb{C}, \; f(z) = \sin z = \frac{e^{iz} - e^{-iz}}{2i}$
4. $f: \mathbb{C} \to \mathbb{C}, \; f(z) = \cos z = \frac{e^{iz} + e^{-iz}}{2}$
5. $f: \mathbb{C} \to \mathbb{C}, \; f(z) = \sinh(z) / \sh(z) = \frac{e^z - e^{-z}}{2}$
6. $f: \mathbb{C} \to \mathbb{C}, \; f(z) = \cosh(z) / \ch(z) = \frac{e^z + e^{-z}}{2}$
7. $f: \mathbb{C}\setminus\{0\} \to \mathbb{C}, \; f(z) = \mathcal{L}n(z) = \ln(z) + i(\theta + 2k\pi)$
8. $f: \mathbb{C} \to \mathbb{C}, \; f(z) = \sqrt[n]{z} = \sqrt[n]{|z|} \left(\cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{2} \right)$
9. $f: \mathbb{C}\setminus\{0\} \to \mathbb{C}, \; f(z) = z^2 = e^{2 \cdot \mathcal{L}n z}$

1. $z=(1-i)^2+\frac{i}{i+1}=\frac{1}{2}-\frac{5i}{2}$

2. $z=ie^{i+2\pi}=i\cdot e^{1}(\underbrace{\cos\pi}_{-1}+i\underbrace{\sin\pi}_{0})=-ie$

3. $z=\sin(\frac{\pi}{3}-i)$

4. $z=ch(1+i\frac{\pi}{2})=\frac{e^{1+i\frac{\pi}{2}}+e^{-1-i\frac{\pi}{2}}}{2}=\frac{e^1(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2})+e^{-1}(\cos\frac{\pi}{2}-i\sin\frac{\pi}{2})}{2}$

$=i(\frac{e-e^{-1}}{2})$

5. $z=2\ln(1-i\sqrt{3})$

$|1-i\sqrt{3}|=\sqrt{4}=2$

~~$\varphi=\arcsin\frac{\sqrt{3}}{2}=\frac{\pi}{3}$~~

~~$\varphi=2\pi-\sin$~~

6. $z=\overline{1+i}\cdot\sqrt[3]{-i}=(1-i)\sqrt[3]{i}$

$|-i|=1$

$-i\to(0,-1)$

$\varphi=\frac{3\pi}{2}$

$=(1-i)\sqrt[3]{1}(\cos\frac{\frac{3\pi}{2}+2k\pi}{3}+i\sin\frac{\frac{3\pi}{2}+2k\pi}{3})=(1-i)(\cos\frac{(4k+3)\pi}{6}+i\sin\frac{(4k+3)\pi}{6}$

$Re(z)=\cos\frac{(4k+3)\pi}{6}+\sin\frac{(4k+3)\pi}{6}$

$Im=\sin\frac{(4k+3)\pi}{6}-\cos\frac{(4k+3)\pi}{6}$

7. $z = i^i = e^{i \cdot \ln i} = e^{i(\ln(1) + i(\frac{\pi}{2} + 2k\pi))}$

8. $z = \frac{e^{i\cdot\frac{\pi}{2}} - 2}{1+2i} = \frac{e^0(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}) - 2}{1+2i} = \frac{i-2}{1+2i} \cdot \frac{1-2i}{1-2i}$

$= \frac{(i-2)(1-2i)}{5} = \frac{i+2-2+4i}{5} = i$

9. $z = ch(i\frac{\pi}{6}) \cdot \cos(\pi - i) = \frac{e^{i\frac{\pi}{6}} - e^{-i\frac{\pi}{6}}}{2} \cdot \frac{e^{1+i\pi} + e^{-1-i\pi}}{2}$

$= e^0(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}) \cdot e^{-1}(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}) \cdot e(\cos\pi + i\sin\pi) + e^{-1}$

$e(\cos\pi + i\sin\pi) - e^{-1}(\cos(-\pi) + i\sin(-\pi))$

$= \frac{\frac{\sqrt{3}}{2} - \frac{i}{2} - \frac{1}{e}(\frac{\sqrt{3}}{2} + i\frac{1}{2})}{2} \cdot \frac{-e + \frac{1}{e}}{2} = \frac{\frac{\sqrt{3}}{2} - \frac{i}{2} - \frac{\sqrt{3}}{2e} + \frac{i}{2e}}{2} \cdot \frac{-e + \frac{1}{e}}{2}$

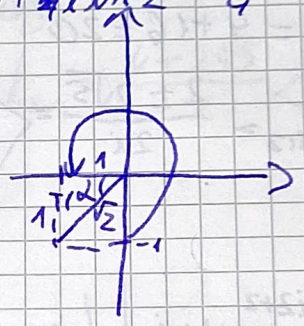
10. $z = (1+2i)\ln(-1-i) = (1+2i)(\ln(\sqrt{2}) + i(\frac{5\pi}{4} + 2k\pi))$

$= \ln(\sqrt{2}) + i(\frac{5\pi}{4} + 2k\pi) + i\ln 2 - \frac{5\pi}{4} - 2k\pi$

$e^{\ln z} = z$

$|-1-i| = \sqrt{2}$

$-1-i \to (-1, -1)$



4) $\theta = 2\pi - \frac{\pi}{4} = \frac{8\pi - \pi}{4} = \frac{7\pi}{4}$

2) $\theta = \frac{3\pi}{2} - \frac{\pi}{4} = \frac{5\pi}{4}$

11. $z = (-i)^{-1} + \sqrt[4]{\frac{1+i\sqrt{3}}{2}} = e^{-i\ln(i)} + \sqrt[4]{1}\left(\cos\frac{\frac{\pi}{3}+2k\pi}{4} + i\sin\frac{\frac{\pi}{3}+2k\pi}{4}\right)$

$\left|\frac{1}{2} + i\frac{\sqrt{3}}{2}\right| = \sqrt{\frac{1}{4}+\frac{3}{4}} = 1$

$\frac{1}{2} + i\frac{\sqrt{3}}{2} \to \theta = \frac{\pi}{3}$

$= e^{-i\left(\ln 1 + i\frac{3\pi}{2} + 2k\pi\right)} + \left(\cos\frac{\frac{\pi}{3}+2k\pi}{4} + i\sin\frac{\frac{\pi}{3}+2k\pi}{4}\right)$

$e^{\frac{3\pi}{2}+2k\pi} + \cos\frac{\frac{\pi}{3}+2k\pi}{4} + i\sin\frac{\frac{\pi}{3}+2k\pi}{4}$

Re Im

$k \in \{0,1,2,3\}.$

Solve in $\mathbb{C}$ the equations.

1. $iz^2 - 2z + 4i = 0$

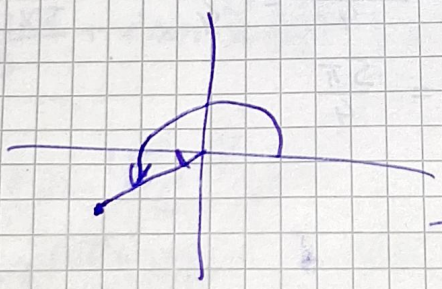
$\Delta = 4 + 16 = 20$

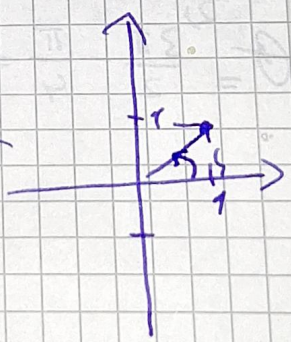
$z_{1,2} = \frac{2 \pm 2\sqrt{5}}{2i} = \begin{cases} \frac{2+2\sqrt{5}}{2i} = \frac{1+\sqrt{5}}{i} = -i + i\sqrt{5} \\ \frac{1-\sqrt{5}}{i} = -i + i\sqrt{5} \end{cases}$

2. $e^{iz+2} = 1+i \mid \ln$

$iz + 2 = \ln(1+i)$

$|1+i| = \sqrt{2}$





$1+i \to (1,1) \to \frac{\pi}{4}$

$iz + 2 = \ln\sqrt{2} + i\left(\frac{\pi}{4} + 2k\pi\right)$

~~$z = \frac{\pi}{4} + 2k\pi$~~ $z = \frac{\ln\sqrt{2} + i\left(\frac{\pi}{4}+2k\pi\right)}{i} = -i\ln\sqrt{2} + \frac{\pi}{4} + 2k\pi$

3. $z^3 = -1-i \Rightarrow e^{3\operatorname{Ln} z} = -1-i \Rightarrow e^3 + 2 = -1-i$

$z = -1-i^3 i$

$x^3 = 7$

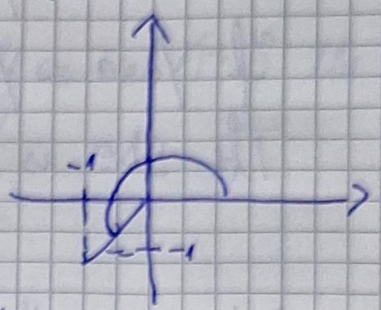
$z = \sqrt[3]{-1-i} = \sqrt[3]{\sqrt{2}}\sqrt[6]{2}\left(\cos\frac{\frac{5\pi}{4}+2k\pi}{3} + i\sin\frac{\frac{5\pi}{4}+2k\pi}{3}\right)$

$|-1-i| = \sqrt{2}$

$k = \{0, 1, 2\}$

$-1-i \to (-1, -1)$

$\theta = \frac{3\pi}{2} - \frac{\pi}{4} = \frac{5\pi}{4}$



4. $\operatorname{sh} z = 2 \Rightarrow \frac{e^z - e^{-z}}{2} = 2 \Rightarrow e^z - e^{-z} = 4 \Rightarrow t - \frac{1}{t} = 4$

Let $e^z = t$

$t^2 - 4t + 1 = 0$

$\Delta = 16 + 4 = 20$

$t_{1,2} = \frac{4 \pm 2\sqrt{5}}{2}$ → $2+\sqrt{5}$, $2-\sqrt{5}$

$e^z = 2+\sqrt{5}$

$t = 2+\sqrt{5} \Rightarrow e^z = 2+\sqrt{5} \mid \operatorname{Ln}$

$z = \operatorname{Ln}(2+\sqrt{5}) = \ln(2+\sqrt{5}) + i(0 + 2k\pi)$

$t = 2-\sqrt{5} \Rightarrow e^z = 2-\sqrt{5} \mid \operatorname{Ln}$

$z = \operatorname{Ln}(2-\sqrt{5}) = \ln(2-\sqrt{5}) + i($

COURSE

Stability

IVP: $Y'(x) = A \cdot Y(x)$ (1') homogeneous SDE, A a real $n \times n$ matrix

$Y(0) = Y_0$ (1C)

$\Rightarrow$ Solution $Y(x) = e^{Ax} \cdot Y_0$

If $Y_0 = \bar{0} \Rightarrow Y(x) = 0 \Rightarrow \bar{0}$ is the equilibrium position

The SDE is asymptotically stable if: - it is stable

- $\lim_{x \to \infty} Y(x) = \bar{0}$

Determine the characteristic polynomial of the matrix A

$P(s) = \det(sI - A) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0$, $a_n > 0$

Determine the eigenvalues of A (the roots of $P(s)$)

$\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$

Theorem:

1. The SDE (1') is asymptotically stable $\Leftrightarrow \forall \lambda \in \sigma(A), \operatorname{Re}\lambda < 0$

a. Routh-Hurwitz Criterion

Notations:

$\mathbb{C}^- = \{z \in \mathbb{C} \mid \operatorname{Re} z = x < 0\}$ (half-plane)

$\mathbb{C}^+ = \{z \in \mathbb{C} \mid \operatorname{Re} z = x > 0\}$

$\mathbb{C}^0 = \{z \in \mathbb{C} \mid \operatorname{Re} z = 0\}$

N^- = the number of the eigenvalues λ of A with $Re\lambda < 0$

N^+ = ——— " ——— " ——— " λ of A with $Re\lambda > 0$

N^0 = ——— " ——— " ——— " λ of A with $Re\lambda = 0$

By Theorem: the SDE (1) is asymptotically stable $\Leftrightarrow$ (all eigenvalues $\in \mathbb{C}^-$) $\Leftrightarrow N^+ = 0,\ N^0 = 0$

We associate to the characteristic polynomial $P(s)$ the Hurwitz matrix

$$H_n = \begin{vmatrix} a_{n-1} & a_n & 0 & 0 & 0 & \dots & 0 & 0 \\ a_{n-3} & a_{n-2} & a_{n-1} & a_n & 0 & \dots & 0 & 0 \\ a_{n-5} & a_{n-4} & a_{n-3} & a_{n-2} & a_{n-1} & a_n & 0 & \dots 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \dots & \dots & a_0 \end{vmatrix}$$

Determine the principle minors of H_n: $\Delta_1, \Delta_2, \Delta_3, \dots$

$\Delta_1 = a_{n-1}$

$$\Delta_2 = \begin{vmatrix} a_{n-1} & a_n \\ a_{n-3} & a_{n-2} \end{vmatrix}$$

$\vdots$

$\Delta_n = \det(H_n)$

Theorem: Routh-Hurwitz Criterion

The SDE is asymptotically stable $\Leftrightarrow \Delta_1 > 0,\ \Delta_2 > 0, \dots, \Delta_n > 0$ (2)

Proof (Gantmacher. Matrix Theory)

Proposition 1: "If $\Delta_i \neq 0$, $\forall i = \overline{1,n}$, then $N^0 = 0$

Proposition 2: N^+ (the number of eigenvalues λ with $\mathrm{Re}\,\lambda > 0$) is equal to the number of changes of signs (N_{CS}) on the sequence: $1, \Delta_1, \frac{\Delta_2}{\Delta_1}, \frac{\Delta_3}{\Delta_2}, \dots, \frac{\Delta_n}{\Delta_{n-1}}$ (3)

The SDE is asymptotically stable $\Leftrightarrow$ (2) holds ($\Delta_1 > 0, \Delta_2 > 0 \dots$)

Sufficiancy: If (2) holds then $\Delta_1 \neq 0, \Delta_2 \neq 0, \dots, \Delta_n \neq 0 \Rightarrow N^0 = 0$

If $\Delta_1 > 0, \Delta_2 > 0, \dots, \Delta_n > 0 \Rightarrow$ all ratios $\frac{\Delta_i}{\Delta_{i-1}} > 0 \Rightarrow$ no change in sign

By Proposition 2: $N^+ = N_{CS} = 0$

$N^0 = 0, N^+ = 0 \Rightarrow$ all eigenvalues have negative real parts

Necessity: Assume the system is asymptotically stable $\Rightarrow$

$\Rightarrow N^0 = 0, N^+ = 0 \Rightarrow N_{CS} = N^+ = 0 \Rightarrow$ no change in sign in (3)

$1 > 0 \Rightarrow \Delta_1 > 0 \Rightarrow \left.\begin{matrix} \frac{\Delta_2}{\Delta_1} > 0 \\ \Delta_1 > 0 \end{matrix}\right\} \Rightarrow \left.\begin{matrix} \frac{\Delta_3}{\Delta_2} > 0 \\ \Delta_2 > 0 \end{matrix}\right\} \Rightarrow \left.\begin{matrix} \frac{\Delta_{n-1}}{\Delta_{n-2}} > 0 \\ \Delta_{n-2} > 0 \end{matrix}\right\}$

§3 Lyapunov Equation

Let P be an $n \times n$ matrix with complex eigenvalues

Definition: The matrix P is positive definite

If $\begin{cases} 1.\ P^* = P \\ 2.\ V^* P V > 0,\ \forall * V \in \mathbb{C}^n,\ V \neq 0 \end{cases}$

A complex number z is real $\Leftrightarrow z = \bar{z}$

z real $\Rightarrow z = x \in \mathbb{R} \Rightarrow z = x + i \cdot 0 \Rightarrow \bar{z} = x - i \cdot 0 \Rightarrow z = \bar{z}$

Conversely, if $z = x + iy \in \mathbb{C}$ verifies $z = \bar{z} \Rightarrow z = x \in \mathbb{R}$

Then if P is a real matrix: $\bar{P} = P \Rightarrow P^* = \bar{P}^T = P^T$

We can prove that a real matrix P is positive definite

$\Leftrightarrow \begin{cases} 1.\ P^T = P \text{ (symetrical)} \\ 2.\ V^T \cdot P \cdot V = 0,\ \forall * V \in \mathbb{R},\ V \neq 0 \end{cases}$

Theorem: The SDE (1') is asimptotically stable $\Leftrightarrow$ the Lyapunov equation ~~PA~~ $PA + A^T P = -Q$ (L), has a positive definite real matrix P for a positive real matrix Q

Proof: Sufficiency $P_1 \Leftarrow P_2$ Assume that Lyapunov equation (L) has a solution $P>0$ for $Q>0$

We will prove that the SDE(1') is asymptotically stable $\forall \lambda \in \sigma(A)$, $\mathrm{Re}\,\lambda < 0$

Let $\lambda \in \mathbb{C}$ be an eigenvalue of the matrix A and $v \in \mathbb{C}^n$ a corresponding eigenvector $\Rightarrow \lambda$ scalar

$$\begin{cases} Av = \lambda v \\ v \neq \bar{0} \end{cases}$$

$$v^* \bar{A} = \bar{\lambda} \cdot v^*$$

Idea: Premultiply (L) by v^* and post multiply by v

$$v^* P A v + v^* A^T P v = -v^* Q v$$

$$\lambda v^* P v + \bar{\lambda} v^* P v = -v^* Q v$$

$$-(\lambda + \bar{\lambda}) v^* P v = -v^* Q v$$

$$2\,\mathrm{Re}\,\lambda\, v^* P v = -v^* Q v$$

$\Rightarrow \mathrm{Re}\,\lambda < 0$, λ eigenvalues of A

Necessity: Assume that the SDE is asymptotically stable $\Rightarrow$
$\Rightarrow \forall \lambda \in \sigma(A), \operatorname{Re}\lambda < 0$

Let $Q > 0$ be a matrix

The solution of $(\star)$ is $P = \int_0^\infty e^{A^T x} Q e^{Ax} dx$

$\operatorname{Re}\lambda < 0 \Rightarrow$ the entries of the matrix P, in modulus have the form $\int_0^\infty x^{\beta-1} e^{-\alpha x} dx$ (Gamma type integral) $\Rightarrow$
$\Rightarrow$ convergent integral $\Rightarrow P$ is well defined

SEMINAR

Functions of a complex variable

$f(z) \overset{z=x+iy}{=\!=} u(x,y) + iv(x,y)$

$Re(f) = u(x,y)$

$Im(f) = v(x,y)$

1. Find the real and imaginary part

$f(z) = i \cdot \sin(z^2) = i\sin((x+iy)^2) = i \cdot \frac{e^{i(x+yi)^2} - e^{-i(x+iy)^2}}{2i}$

$= \frac{e^{i(x^2-y^2+2ixy)} - e^{-i(x^2-y^2+2ixy)}}{2} = e^{-2xy}(\cos(x^2-y^2) + i\sin(x^2-y^2)) -$

$e^{2xy}(\cos(x^2-y^2) + i\sin(x^2-y^2))$

2. $f(z) = \frac{e^{iz-1}}{z} = \frac{e^{-y+ix-1}}{x+iy} = \frac{e^x(\cos(-y-1) + i\sin(-y-1))}{x+iy}$

$x-iy$

$= \frac{e^x(\cos(y+1) - i\sin(-y-1)}{x+iy} = \frac{(x-iy)(e^x(\cos(y+1) + i\sin(-y-1)}{x^2+y^2}$

$= \frac{(x-iy)(e^x\cos(y+1) + ie^x\sin(-y-1)}{x^2+y^2} =$

$= \frac{xe^x\cos(y+1) + ye^x\sin(-y-1)}{x^2+y^2} + \frac{-iye^x\cos(y+1) + xie^x\sin(-y-1)}{x^2+y^2}$

$$Re(f) = \frac{e^x[x\cos(y+1) - y\sin(y+1)]}{x^2+y^2}$$

$$Im(f) = \frac{e^x[-x\sin(y+1) - y\cos(y+1)]}{x^2+y^2}$$

$e^{-y-1} \cdot (\cos x + i,$

Checking the derivability of f at $z_0 = x_0 + iy_0$, $x_0, y_0 \in \mathbb{R}$

Algorithm:

1. Replace $z = x + iy$ and find $\mathrm{Re}(f) = u(x,y)$ and $\mathrm{Im}(f) = v(x,y)$
2. Compute $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$
3. Check if Cauchy-Riemann conditions are true:

a. $\frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0)$

b. $\frac{\partial u}{\partial y}(x_0, y_0) + \frac{\partial v}{\partial x}(x_0, y_0) = 0$

If both are true $\Rightarrow$ f is derivable at z_0

Otherwise, f isn't derivable at z_0

3. Check derivability of $f(z)$ at z_0

$f(z) = (\bar{z} - i)e^{\bar{z}}$, $z_0 = -i$

$f(z) = (x - iy - i)e^{x-iy} = (x - i(y+i)) \cdot e^{x-iy} = (x - i(y+i))$

$e^x(\cos y + i \sin y) = e^x(x\cos y - xi\sin y + i(y+i)\cos y - (y+i)\sin$

$\mathrm{Re}(f) = e^x[x\cos y - (y+1)\sin y]$

$\mathrm{Im}(f) = e^x[-x\sin y - (y+1)\cos y]$

$\frac{\partial u}{\partial x} = e^x[x\cos y - (y+1)\sin y] + e^x \cos y$

$\frac{\partial u}{\partial y} = e^x[-x\sin y - \sin y - (y+1)\cos y]$

$$\frac{\partial v}{\partial x} = e^x[-x\sin y - (y+1)\cos y] + e^x(-\sin y)$$

$$\frac{\partial v}{\partial y} = e^x[-x\cos y - \cos y + (y+1)\sin y]$$

$$\frac{\partial u}{\partial x}(0,-1) = 2\cos 1$$

$$\frac{\partial v}{\partial y}(0,-1) = -\cos 1$$

4. $f(z) = \frac{i}{z} - ch z$, $z_0 = -1-i$

$$f(z) = \frac{i}{x+iy} - \frac{e^{x+iy} + e^{-x-iy}}{2} = \frac{x-iy}{x^2+y^2} - e^x(\cos(y)+i\sin y) + e^{-x}(\cos(-y)+i\sin(-y))$$

$$= \frac{(x-iy)x - (x^2+y^2)[e^x(\cos(y)+i\sin(y)) + e^{-x}\{\cos(-y)+i\sin(-y)\}]}{2(x^2+y^2)}$$

$$= \frac{2x + 2iy - x^2e^x\cos y - x^2ie^x\sin(y) + e^{-x}x^2\cos(-y) - x^2e^{-x}i\sin(-y) + y^2e^x\cos(y) + y^2ie^x\sin(y) + y^2e^{-x}\cos(-y) + y^2ie^{-x}\sin(-y)}{2(x^2+y^2)}$$

$$Re(f) = \frac{2x - x^2e^x\cos y - e^{-x}x^2\cos(-y) + y^2e^x\cos(y) + y^2e^{-x}\cos(-y)}{2(x^2+y^2)}$$

$$Im(f) = \frac{2y - x^2e^x\sin(y) - x^2e^{-x}i\sin(-y) + y^2e^x\sin(y) + y^2e^{-x}\sin(-y)}{2(x^2+y^2)}$$

$$Re(f) = \frac{x}{x^2+y^2} - \cos y \cdot ch x$$

$$Im(f) = \frac{y}{x^2+y^2} - \sin y \cdot sh x$$

$$\frac{\partial u}{\partial x} = \frac{-2xy}{(x^2+y^2)^2} - \cos y \operatorname{sh} x \quad \checkmark$$

$$\frac{\partial u}{\partial y} = \frac{x^2+y^2-2y^2}{(x^2+y^2)^2} + \sin y \operatorname{ch} x \qquad \frac{(x^2+y^2-2y^2}{(x^2+y^2)^2} + \sin y \operatorname{ch}$$

$$\frac{\partial v}{\partial x} = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} - \sin y \operatorname{ch} x \quad \checkmark$$

$$\frac{\partial v}{\partial y} = \frac{-2xy}{(x^2+y^2)} - \cos y \operatorname{sh} x \quad \checkmark$$

$$\frac{\partial u}{\partial x}(-1,-1) = -\frac{1}{2} + \cos 1 \operatorname{sh} 1$$

$$\frac{\partial v}{\partial y}(-1,-1) = -\frac{1}{2} + \cos 1 \operatorname{sh} 1$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \Rightarrow f \text{ derivable at } z_0$$

5. $f(z) = \cos(z - i\bar{z})$, $z_0 = 2 + 2i$

$$f(z) = \cos(x + iy - i(x - iy)) = \frac{e^{i(x-y+iy-ix)} + e^{-i(x-y+iy-ix)}}{2}$$

$$= \frac{e^{(x-y)+i(x+y)} + e^{(y-x)+i(-x+y)}}{2} =$$

$$= \frac{e^{x-y}(\cos(x-y) + i\sin(x-y)) + e^{y-x}(\cos(y-x) + i\sin(y-x))}{2}$$

$$\text{Re } u(x,y) = \frac{e^{x-y}\cos(x-y) + e^{y-x}\cos(y-x)}{2} = \frac{\cos(x-y)(e^{x-y} + e^{y-x})}{2}$$

$$V(x,y) = \frac{1}{2}(e^{x-y}\sin(x-y) + e^{y-x}\sin(y-x)) = \frac{\sin(x-y)(e^{x-y} - e^{y-x})}{2}$$

$$\cos(y-x) = \cos(x-y)$$

$$\sin(y-x) = -\sin(x-y)$$

$$u(x,y) = \cos(x-y) \cdot \operatorname{ch}(x-y)$$

$$V(x,y) = \sin(x-y)\operatorname{sh}(x-y)$$

$$\frac{e^{x-y} + e^{y-x}}{2} = \operatorname{ch}(x-y)$$

$$\frac{e^{x-y} - e^{y-x}}{2} = \operatorname{sh}(x-y)$$

$$\frac{\partial u}{\partial x} = -\sin(x-y)\cdot\operatorname{sh}(x-y) - \sin(x-y)\cdot\operatorname{ch}(x-y) + \cos(x-y)\cdot\operatorname{sh}(x-y)$$

$$\frac{\partial u}{\partial y} = -\sin(x-y)\cdot$$

6. $f(z) = (\bar{z}+1)e^{\bar{z}}$, $z_0 = 0$

$f(z) = (x+1-iy)e^{x-iy} = (x+1-iy)e^x(\cos(-y)+i\sin(-y))$.

$= (x+1-iy)e^x(\cos y - i\sin y) = xe^x\cos y - ixe^x\sin y + e^x\cos y - ie^x\sin y - iye^x\cos y - ye^x\sin y$

$u(x,y) = e^x(x\cos y + \cos y - y\sin y)$

$v(x,y) = e^x(-x\sin y - \sin y - y\cos y)$

$\bar{z}e^{\bar{z}} + e^{\bar{z}}$

$\frac{\partial u}{\partial x} = e^x(\cos y + \cos y - y\sin y) + e^x\cdot\cos y$

$\frac{\partial u}{\partial y} = e^x(-x\sin y - \sin y + \sin y + y\cos y)$

$\frac{\partial v}{\partial x} = e^x(-x\sin y - \sin y - y\cos y) = e^x\sin y$

$\frac{\partial v}{\partial y} = e^x(-x\cos y - \cos y - \cos y + y\sin y)$

$\left.\begin{array}{l}\frac{\partial u}{\partial x}(0,0) = 2 \\ \frac{\partial v}{\partial y} = -2\end{array}\right\} \Rightarrow$ CRI not fullfield $\Rightarrow f$ not derivable in z_0